DANE MALENFANT

MACHINE LEARNING RESEARCH / POST-TRAINING / EVALUATIONS / REINFORCEMENT LEARNING

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RESEARCH PROFILE

Machine learning researcher studying how post-training changes agent behavior beyond aggregate benchmark scores. Sole-authored research compares SFT, RLOO and prompt optimization using diagnostic evaluations and counterfactual credit assignment; co-authored work studies scalable memory for long-horizon agentic RL. MSc Computer Science, McGill / Mila.

SELECTED RESEARCH & EXPERIMENTAL CONTRIBUTIONS

Language-model post-training, evaluations & credit assignment | Sole-author research, 2026

- Built the Dialogue Moral Hazard Game, a controlled task for costly safety-information sharing. Evaluated 14 open-weight and 4 frontier models, including matched 3,015-decision-per-model incentive sweeps.
- **Compared post-training recipes** using multi-seed LoRA SFT, RLOO and sequential SFT+RLOO; studied GEPA prompt optimization with held-out evaluation, frozen episodes, baselines and ablations.
- **Diagnosed benchmark shortcuts:** For Muse Spark 1.1, GEPA raised team success from 22.2% to 100% while query use fell from 51.1% to 0.3%; rank-label interventions revealed brittle learned behavior.
- **Introduced CREDIT,** a counterfactual-replay prompt-optimization method using matched hidden-state twins to reward evidence-driven information transfer; studied acquisition and downstream-use bottlenecks across 5 models and multiple seeds.

Multi-agent reinforcement learning & policy gradients | RLC 2026

- Co-developed a hidden-gifts grid world, compared multi-agent RL algorithms, and derived a learning-aware policy-gradient correction that reduced variance and improved collective success.

Continual-learning reliability & registered experiments | 2026 preprint

- Analyzed stability of success-conditioned structure under peer-policy drift; designed two registered 64-stream studies (continual control and cue-MNIST) in which structural erosion predicted failure and supported near-oracle intervention.

Agentic RL training efficiency & recurrent memory | KV-streams, arXiv 2026

- Co-authored KV-streams, a KV-cache-preserving compaction method for long-horizon agentic RL. The study reports 2.6–5× faster training without evidence of performance loss, and shows recurrent memory can emerge from RL alone.

TECHNICAL & EXPERIMENTAL METHODS

Engineering Python · PyTorch · NumPy · vLLM · Slurm · Git · W&B · local and hosted LLM evaluation · tested research software

Training & eval LoRA/SFT · RLOO · GEPA prompt optimization · RL policy gradients · paired/multi-seed sweeps · ablations · frozen evaluation data · decomposed metrics

Compute GPU training and five seeded Slurm jobs (GFlowNet course project); computational support from DRAC, Mila IDT and NERSC

EDUCATION

2026 **MSc, Computer Science** | McGill University; thesis research at Mila (LiNC lab, Blake Richards). Thesis approved May; final submitted August 2026.

2022 **Bachelor's degree, Computer Science** | McGill University; additional studies in psychology.

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PROFESSIONAL EXPERIENCE

2026 **AI Engineering & Software Intern** | Veris AI · New York; began August 2026

2023–24 **Consultant** | AI for Humanity, Mila · Indigenous Pathfinders in AI and FLAIR, Montréal

2023–24 **Interim Program Manager / Assistant Program Coordinator** | Branches Program, McGill University

2022–23 **Research Assistant & Research Intern** | Mila / The Neuro · LiNC lab, Montréal

2021 **Cybersecurity Analyst** | RBC · Remote / Toronto

SELECTED PUBLICATIONS

2026 **Moral Hazard in Multi-Agent Language Models**
D. Malenfant · arXiv:2607.23982 (revised September 2026)

2026 **KV-streams for Efficient Compaction in Agentic Reinforcement Learning**
E. Penaloza, D. Malenfant, D. Vattikonda et al. · arXiv:2609.35750 (preprint)

2026 **The Challenge of Hidden Gifts in Multi-Agent Reinforcement Learning**
D. Malenfant and B. A. Richards · Reinforcement Learning Journal 7 / RLC 2026

2026 **Reinforcing the World's Edge: Continual Learning at the Multi-Agent–World Boundary**
D. Malenfant · arXiv:2603.06813; World Modeling Workshop 2026

2026 **Compositional Flow Matching with Factored Velocity Fields**
A. H.-W. Ryoo, D. Malenfant, M. G. Perich and G. Lajoie · FoGen at ICML 2026 (spotlight)

2024 **Inconsistencies in Artificial Intelligence Strategy Alignment of NATO Member States**
I. Epstein, D. Malenfant, S. Parker and C. Wardrop · Harms and Risks of AI in the Military

2023 **Contrastive Retrospection: Honing in on Critical Steps for Rapid Learning and Generalization in RL**
C. Sun, W. Yang, T. Jiralerspong, D. Malenfant et al. · NeurIPS 36

2022 **Causality Mining in Fiction**
M. Meehan, D. Malenfant and A. Piper · Text2Story at ECIR 2022

OPEN RESEARCH SOFTWARE

Dialogue Moral Hazard Game · MIT-licensed Python task and eval package: frozen episodes, baselines, tests, decomposed metrics, local vLLM / hosted API sweeps.

Reinforcing the World's Edge · MIT-licensed experiment code, registered 64-stream results, structural-law tests and figure-reproduction scripts.

HONORS & RESEARCH COMMUNITY

Honors Tiplady Sweeney Fellowship (2026); NSERC Canada Graduate Scholarship - Master's and Rathlyn Fellowship (2024); Best Poster awards (HBHL 2025; Montréal AI and Neuroscience 2024).

Service Organizer, Context Beyond the Window at COLM 2026; official reviewer for ICLR, NeurIPS, European Workshop on Reinforcement Learning and workshops (2024–26).

Mentorship Supervised COGS 401/444 research on emergent reciprocity (2024–25); advisory board, Indigenous Pathfinders in AI.

French DELF B2; University of Regina certificate in French as a Second Language (2018).